Letter Sequences & Alphabet Code (Set B)

Year 4 · **Homework Worksheet**

Worked Example

Example — Mixed Code Practice: What comes next? C, F, I, L, __ and if A=1, B=2..., what number represents the letter in your answer?

Answer: O comes next (gap of +3 throughout: C, F, I, L, O). O is the 15th letter, so the number is 15.

Quick Practice (15 questions)

1. What comes next? B, D, F, H, __
2. What comes next? W, U, S, Q, __
3. If A=1, B=2, C=3..., what number stands for the letter J? A) 9 B) 10 C) 11 D) 12 E) 8
4. Select ALL that apply: which of these pairs of letters are exactly 3 letters apart? A) B and E B) F and I C) K and L D) N and Q E) R and S
5. What comes next? A, D, G, J, __
6. In a code where A=1, B=2, C=3..., which letter stands for the number 20? A) R B) S C) T D) U E) V
7. What comes next? Y, V, S, P, __
8. Fill in the missing letter: B, _, F, H, J (skipping one letter each time)
9. If every letter is shifted forward by 2 places (A→C, B→D...), what does E become? A) C B) D C) F D) G E) H
10. Select ALL that apply: in the sequence D, G, J, M, P, which statements are true? A) The gap between each letter is 3. B) The sequence decreases. C) The next letter after P is S. D) The first letter is D. E) The gap between J and M is 4.
11. What comes next? E, G, I, K, __
12. In the alphabet-position code (A=1...), what number represents the letter N? A) 13 B) 14 C) 15 D) 12 E) 16
13. What comes next? T, R, P, N, __
14. If letters are shifted BACK by 1 place (B→A, C→B...), what does G become? A) E B) F C) H D) I E) D
15. Select ALL that apply: which letters come BEFORE J in the alphabet? A) G B) M C) C D) L E) F

Mixed Practice (15 questions)

1. What comes next? A, C, F, J, __ (the gap increases by 1 each time: +2, +3, +4...)
2. A code shifts every letter forward by 3 places (A→D, B→E...). What does the word "DOG" become?
3. What comes next? B, E, I, N, __ (gaps increase by 1 each time: +3, +4, +5...)
4. In a code, A=1, B=2... Two numbers are added and converted to a letter: if the numbers are 9 and 10, which letter results? A) R B) S C) T D) U E) Q
5. Select ALL that apply: which sequences correctly keep a GAP of 5 between each letter? A) A, F, K, P, U B) B, G, L, Q, V C) C, H, M, R, X D) D, I, N, S, X E) A, E, I, M, Q
6. What comes next? Y, V, S, P, __ (gap is always -3) — give the next TWO letters.
7. A code shifts every letter BACK by 4 places (E→A, F→B...). What does the word "JQM" decode to?

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8. What comes next in this interleaved sequence? A, Y, C, W, E, __

9. In the position code (A=1…Z=26), what letter corresponds to 26 − 10? A) O B) P C) Q D) N E) R

10. Select ALL that apply: in a +6 shift code (A→G, B→H…), which of these are correctly matched? A) A→G B) D→J C) S→Y D) U→A (wrapping) E) Z→F (wrapping)

11. What comes next? D, G, K, P, __ (gap increases by 1 each time: +3, +4, +5…)

12. A code uses "position number + 4, then convert back to a letter" (wrap after Z). What does the letter X become?

13. What comes next? N, K, H, E, __

14. Select ALL that apply: in the sequence F, I, L, O, R, which of the following are correct? A) Each gap is +3. B) The sequence is decreasing. C) The next letter after R is U. D) F is the first term. E) The gap between L and O is 2.

15. In a shift code where every letter moves forward by 6 places, what does the letter N become?

Standard / Advanced (15 questions)

1. What letter comes immediately after the next term in the sequence: D, G, J, M, ?

2. If the code for FOX is HQZ, what letter would C become using the same code?

3. What letter comes immediately after the next term in the sequence: X, U, R, O, ?

4. If the code for HEN is KHQ, what letter would B become using the same code? A) C B) D C) E D) F E) G

5. In a position code (A=1…Z=26), what letter is 2 places after the letter whose position number is 5?

6. A pattern of letters moves the same number of places forward through the alphabet each time. The first 4 letters are C, G, K, O. What letter comes two terms further along the pattern after O?

7. A pattern of letters moves the same number of places back through the alphabet each time. The first 4 letters are V, S, P, M. What letter comes two terms further along the pattern after M?

8. A tutor makes up a code: every letter of a word is moved 3 places forward in the alphabet. Using this code, what coded word would be made from the word CAT? A) FDW B) FDV C) EDW D) FCW E) FEW

9. A tutor makes up a code: every letter of a word is moved 2 places forward in the alphabet. Using this code, what coded word would be made from the word FISH?

10. A message has been coded by moving every letter 2 places forward in the alphabet. The coded word is "ECV". What was the original word before it was coded?

11. Select ALL that apply: a code shifts every letter forward by 2 places, and under this code DOG becomes FQI. A) The shift used is +2. B) Using the same shift, CAT would become ECV. C) Using the same shift, CAT would become EBV. D) The shift used is -2. E) Using the same shift, SUN would become UWP.

12. In a shift code, the letter C becomes F. Using the same shift, what would the letter L become?

13. A puzzle gives you this pair as a worked example of a code: LEG → NGI. Work out the shift, then encode the word "CAT" using the same shift. A) ECV B) EBV C) DCV D) ECW E) FCV

14. A puzzle gives you this pair as a worked example of a code: BIRD → DKTF. Work out the shift, then decode the word "NGUV" using the same shift. A) LEST B) LESS C) LEAT D) MEST E) LFST

15. A two-step code first shifts every letter back by 1 place, then swaps the first and last letters of the result. Apply this to the word "DOG" and give the final 3-letter result.

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